Identifiability and Experimental Design for Neural-Mechanistic Hybrid Models Using Physics-Informed Neural Networks in Colorectal Cancer

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Abstract

Neural mechanistic hybrid models learn unknown regulation while preserving biology, but can obscure parameter values. We use physics informed neural networks in an ordinary differential equation model of WNT, RA, and HOX regulation across four APC loss regimes. Fisher information, profile likelihood, and Hamiltonian Monte Carlo show that beta-catenin saturation makes APC function practically nonidentifiable at high WNT. The network absorbs basal production despite bounded, monotone constraints. A knockdown reaching zero resolves this ambiguity, reducing error from 2.2% to 0.0%, while an information matched control fails. Reachability analysis identifies the least costly knockdown per learned term, and prospective testing confirms predictions. These results support experimental designs that improve the identification of regulatory mechanisms involved in colorectal cancer progression.

01 WNT–RA–HOX model

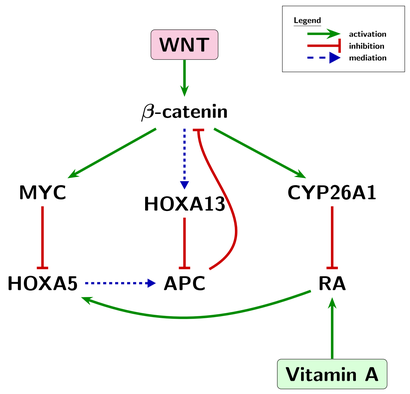


Fig. 1. Regulatory network.

Disease regimes

Regime	W	θ_P
Normal	0.8	1.00
Early adenoma	1.0	0.75
Advanced adenoma	1.5	0.50
Severe APC loss	2.0	0.25

W: WNT drive.
θ_P: APC functioning.

Seven states

$\mathbf{z} = (b, p, h_5, h_{13}, m, r, c)$
b: β-catenin; p: APC
h_5, h_{13}: HOXA5, HOXA13
m: MYC; r: RA; c: CYP26A1

Nondimensional model

Scale concentrations by characteristic values and time by β-catenin turnover, $d_B = 1\,\mathrm{h}^{-1}$:

$$X = X_* x, \qquad \tau = d_B t, \qquad \varepsilon_X = \frac{d_B}{d_X}. \tag{1}$$

Hill activation: $H_n(u;\kappa) = u^n/(\kappa^n + u^n)$. A dot below denotes $d/d\tau$.

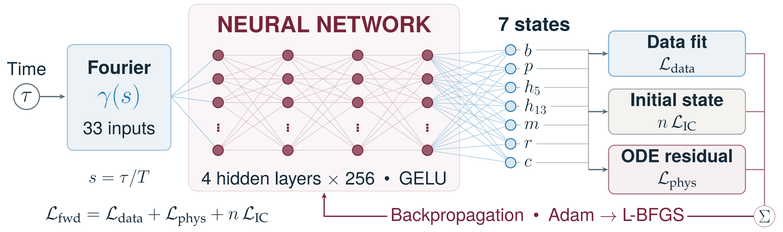
$$\begin{aligned}
\dot{b} &= W + \eta_{13} H_{n_H}(h_{13};\kappa_{13}) - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b},\\
\varepsilon_P \dot{p} &= \frac{1+\rho_5 h_5}{1+\rho_B b + \rho_{13} h_{13}} - D_P(\theta_P) p,\\
\varepsilon_5 \dot{h}_5 &= a_5 + \eta_R H_1(r;\kappa_R) - h_5 - \eta_M \frac{m h_5}{\kappa_M + m},\\
\varepsilon_{13} \dot{h}_{13} &= a_{13} + \eta_{B13} H_{n_B}(b;\kappa_{B13}) + \eta_{M13} H_{n_M}(m;\kappa_{M13}) - h_{13},\\
\varepsilon_M \dot{m} &= a_M + \eta_{BM} H_{n_B}(b;\kappa_{BM}) - m,\\
\varepsilon_R \dot{r} &= \mu_R(\tau) - r - \lambda_C c r,\\
\varepsilon_C \dot{c} &= a_C + \eta_{RC} H_1(r;\kappa_{RC}) + \eta_{BC} H_{n_B}(b;\kappa_{BC}) - c.
\end{aligned} \tag{2}$$

APC functioning & retinoid input

$$\begin{aligned}
D_P(\theta_P) &= 1 + \delta_{P1}(1-\theta_P),\\
\mu_R(\tau) &= \mu_0 + A_R\left[1 + \cos\left(\frac{2\pi\tau}{T_R} - \varphi\right)\right]\\
&\quad + \frac{D_R}{2}\left[\tanh q(\tau-\tau_1) - \tanh q(\tau-\tau_2)\right].
\end{aligned} \tag{3}$$

Radau reference; $\tau \in [0, 150]$. ATRA pulse: $40 \le \tau \le 88$.

02 PINNs & identifiability



$$\mathcal{L}_{\text{phys}} = \frac{1}{N_c}\sum_k \left\| \partial_\tau \widehat{\mathbf{z}}_k - \mathbf{F}(\tau_k, \widehat{\mathbf{z}}_k; \boldsymbol{\theta}) \right\|^2. \tag{4}$$

Forward PINN,[1] implying IC $n = 20$.

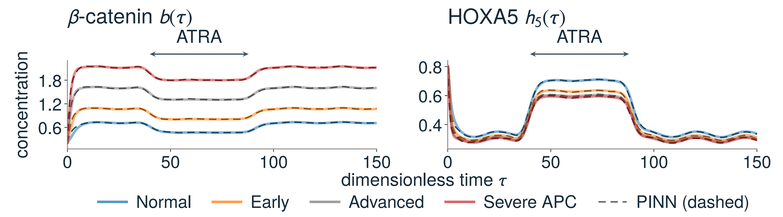


Fig. 2. Forward PINN solution of the ODE system, four regimes.

Inverse PINN: recovering 36 parameters

$$\mathbf{r}_k = \widehat{\mathbf{z}}_{k+1} - \widehat{\mathbf{z}}_k - \tfrac{1}{2}\Delta\tau_k\left(\mathbf{F}_k + \mathbf{F}_{k+1}\right). \tag{5}$$

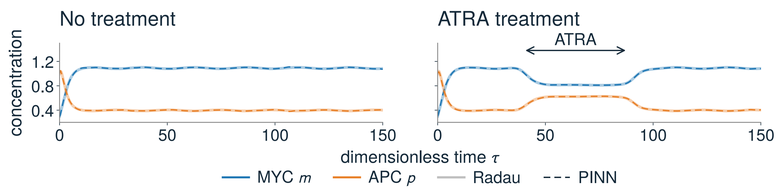
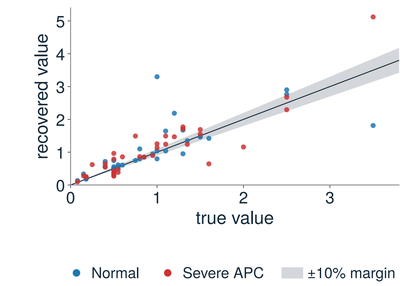


Fig. 3. Inverse PINN fit of Normal regime depending on treatment.

Method	Normal	Early	Advanced	Severe	Total
Autodiff	16	9	6	6	37
Integral	17	16	10	7	50

Table 1. Parameters within 10% error margin.

Recovered vs. true parameters



Local Fisher counts

	I	W	N
Normal	13	22	1
Early	17	18	1
Advanced	14	20	2
Severe	14	21	1

I: identifiable
W: weak
N: non-identifiable

Fig. 4. 36 parameters recovery results for inverse PINN.

03 Neural–mechanistic hybrid

$$\begin{aligned}
\eta_{BM} H_{n_B}(b;\kappa_{BM}) &\longrightarrow f_\phi(b), \qquad f_\phi(0) = 0,\\
\varepsilon_M \dot{m} &= a_M + f_\phi(b) - m.
\end{aligned} \tag{6}$$

f_ϕ: shared 2×5 tanh network.[2] Joint integral fit; L2 penalty 10^{-8}.
Replace (η_{BM}, κ_{BM}): 36 → 34 inverse parameters.

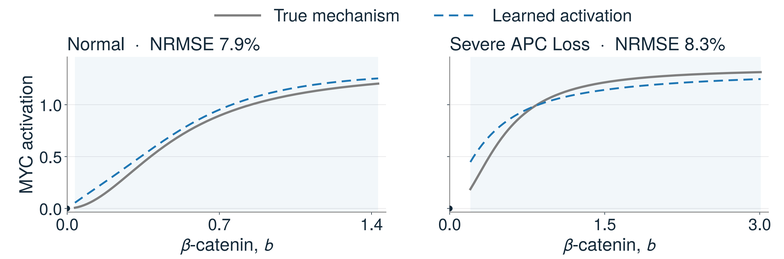


Fig. 5. Learned MYC activation; dots mark $f_\phi(0) = 0$.

Parameter & function recovery

Learned term	Control	Hybrid	NRMSE (%)
RA → HOXA5	51/136	53/136	7.6–23.7
β-catenin → MYC	48/136	43/136	6.2–9.5
APC loss	52/140	53/140	0.12

Table 2. Parameters within 10% error margin, 4 regimes.

Basal–neural compensation[3]

$$a_M + f_\phi(b) = (a_M + c) + \left[f_\phi(b) - c\right]. \tag{7}$$

Designed knockdowns

Basal	Knockdown	Normal	Severe
a_M	WNT depletion	2.17	1.55
a_M	WNT + HOXA13 suppression	0.02	0.01
a_{13}	none	12.84	95.56
a_{13}	β-catenin depletion	51.52	59.84
a_{13}	MYC depletion	0.11	0.05

Table 3. Hybrid basal-parameter error (%) under each knockdown.

Mechanistic vs. hybrid errors

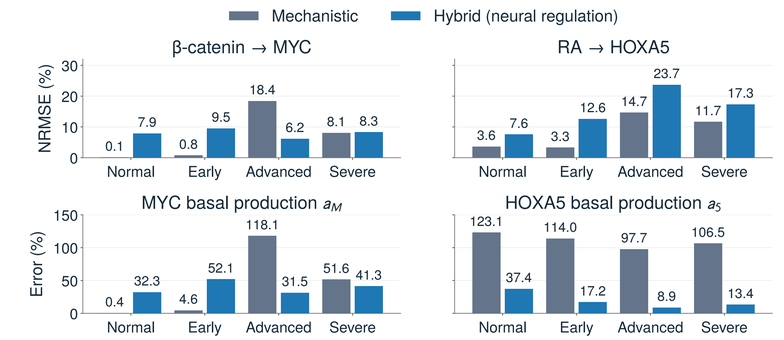


Fig. 6. Mechanistic vs. hybrid errors, full inverse-PINN fits.

References [1] Raissi et al. *J. Comput. Phys.* 378 (2019). doi:10.1016/j.jcp.2018.10.045. [2] Rackauckas et al. Universal differential equations. arXiv:2001.04385 (2020). [3] Loman & Baker. Functional and parametric identifiability for UDEs. arXiv:2510.14140 (2025).